PQC DAILY STUDY SERIES

PQC in the Linux Kernel: Signing Modules with ML-DSA

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Introduction

The Linux kernel, the core of the Linux operating system, is undergoing a significant security enhancement to protect against the threat of quantum computers. As of late 2026, developers are actively working on integrating Post-Quantum Cryptography (PQC) algorithms, specifically **ML-DSA (Module-Lattice-based Digital Signature Algorithm)**, for signing kernel modules. This is a critical step in securing the entire software supply chain, ensuring that the code running in the most privileged level of the operating system is authentic and untampered with.

Why ML-DSA for Kernel Module Signing?

Kernel modules are pieces of code that can be loaded and unloaded into the kernel on demand. They extend the functionality of the kernel without the need to reboot the system. Because they run with kernel-level privileges, their integrity is paramount. An attacker who could load a malicious module would have complete control over the system.

Currently, kernel modules are signed using classical algorithms like RSA or ECDSA. While secure today, these algorithms are vulnerable to attacks from future fault-tolerant quantum computers. ML-DSA, standardized by NIST as FIPS 204, is designed to resist such attacks.

The Integration Process

The integration of ML-DSA into the Linux kernel is a multi-step process:

- **Cryptographic Library Integration:** The core ML-DSA implementation is added to the kernel's crypto library.
- **Signature Verification:** The kernel's module loader is modified to recognize and verify ML-DSA signatures.
- **Toolchain Support:** The tools used to sign kernel modules are updated to support ML-DSA.

This phased approach allows for testing and validation at each stage, ensuring a smooth and secure transition.

Performance Considerations

A key concern with PQC algorithms is their performance. ML-DSA, however, offers a good balance of security and performance, with signature verification times that are comparable to classical algorithms. This is crucial for the kernel, where performance is critical.

Algorithm	Public Key Size (bytes)	Signature Size (bytes)	Security Level (NIST)
ML-DSA-44	1312	2420	2
ML-DSA-65	1952	3293	3
ML-DSA-87	2592	4595	5

Applicability to Real-World Cybersecurity

The integration of ML-DSA into the Linux kernel has far-reaching implications for cybersecurity:

- **Secure Boot:** It strengthens the secure boot process by ensuring that only trusted kernel modules are loaded.
- **IoT and Embedded Devices:** Many IoT and embedded devices run Linux. Securing their kernels with PQC is essential to protect them from remote attacks.
- **Cloud Computing:** Cloud infrastructure relies heavily on Linux. PQC in the kernel protects the foundation of the cloud from quantum threats.

Daily Quiz

1. What is the primary motivation for integrating ML-DSA into the Linux kernel?
2. Why is the performance of ML-DSA important for kernel module signing?
3. Besides the kernel itself, what other parts of the ecosystem need to be updated to support ML-DSA?